

Demo: What the Coil Can Do

Jacob's ladder, and lighting a tube from your hand

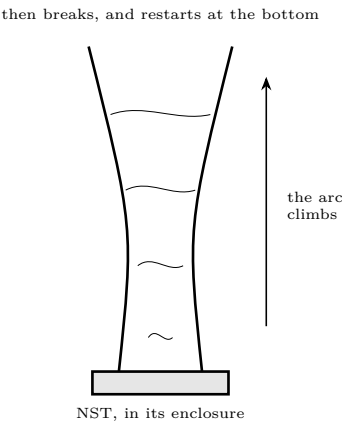
The payoff demonstrations —
adult-operated

Lesson Demo E · after the coil is built · adult owns the mains side

ADULT-OPERATED: MAINS-DERIVED HIGH VOLTAGE

From here on the apparatus contains voltages and currents that kill people every year. The adult owns the mains plug, the transformer and every decision about energizing. Nobody touches anything inside the enclosure until the supply is unplugged *and* the capacitor has been shorted with the grounding stick. Read the **Safety** sheet completely before this lesson, not during it.

Jacob's ladder



Predict first	
Two wires make a narrow V, closest together at the bottom. An arc strikes across the gap and then travels <i>upward</i> , into a wider and wider gap that ought to be harder to jump. Why does it climb rather than staying where it started?	
Why it strikes low	The gap is narrowest at the bottom, so that is where the field is strongest and the air breaks down first. <i>see the 10. The Spark Gap sheet</i>
Why it climbs	The arc is a column of hot ionised gas, and hot gas rises. It also carries current, so it is pushed by its own magnetic field. Both effects drive it up the electrodes.
Why it survives	Ionised air conducts far better than cold air, so once an arc exists it can be stretched much wider than the gap it could originally jump.
Why it breaks	Eventually the arc is stretched too far, cools, and stops conducting. The voltage then reappears across the narrow bottom gap and the cycle restarts.
Build note	Two stiff bare wires, 1–2 mm apart at the base, opening to 5–8 cm at the top, mounted on a fireproof insulating base inside the enclosure.

Lighting a fluorescent tube from your hand

What happens	Stand several feet from a running coil holding an ordinary fluorescent tube, and it lights — with no wires and nothing connected to it.
Why	The coil floods the room with a strong, rapidly changing electric field. That field drives the free electrons in the tube's low-pressure mercury vapour back and forth, they collide with mercury atoms, the atoms emit ultraviolet, and the phosphor coating converts that to visible light. Exactly the same process as in the ceiling fixture, with the field arriving through the air instead of through a ballast.
It is a field meter	Walk toward the coil and it brightens; step behind a grounded metal sheet and it dims. You are mapping the field you drew with iron filings in Lesson 2, at room scale.
Distance rule	Do this at a stated distance, agreed before the coil is switched on, and never with any part of you inside the streamer range.
Not a party trick	Everyone in the room needs to be beyond the longest streamer you have seen from this coil, plus a wide margin. The tube does not have to be close to work.

THESE TWO ARE THE REWARD, NOT THE POINT

Both demonstrations use the mains-derived high-voltage supply, and the full **Safety** sheet applies without exception: named adult on the plug, grounding stick in reach, capacitor shorted before any hand goes near anything, second person present, everyone with an implant out of the room, all electronics elsewhere, eye and ear protection, extinguisher in reach. Run in short bursts. Stop at the first sign of anything unexpected — a smell of hot varnish, an arc where there should not be one, a transformer getting warm.

Close the loop

Ask, before you switch anything on, which lesson each of these comes from. The ladder is Lesson 10 — air breaking down and becoming a conductor. The tube is Lesson 2 — a field reaching across empty space to do work. If they can name those without help, the course did its job, and the sparks are just the receipt.

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